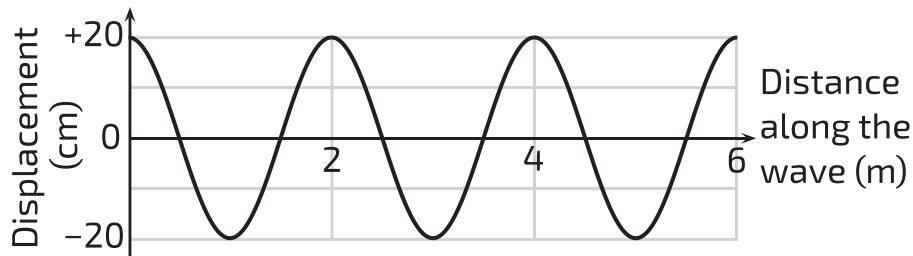


Waves Practice

- 1 Two friends stretch a long skipping rope between them. One of them starts swinging it up and down. The diagram shows what the rope looks like a short time later.

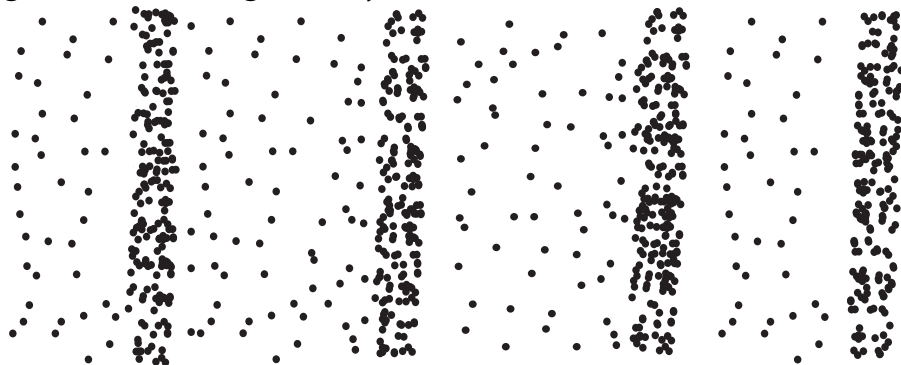


(a) Is the wave transverse or longitudinal?

(b) Using the letters P and T, label all the peaks and troughs on the diagram.

- 2 The diagram shows the particles in a sound wave (longitudinal) passing through a dust cloud.

On the diagram, mark a length that you would measure to measure one wavelength.



- 3 Complete the sentences using the words or phrases **L**, **oscillation**, **trough**, **peak**, **corresponding**.

To measure the wavelength (λ) of a wave, you need to measure from one point to the next _____ point. This ensures that exactly one full cycle has played out across the gap. For example, from peak to _____, or trough to _____.

If you measure horizontally from one peak to the next _____, you will only have measured half the wavelength, because only half a wave-cycle (_____) has occurred.

We use λ as the symbol for **wavelength** because it is the greek letter _____.

- 4 Complete the sentences using the words or phrases **energy**, **oscillation**, **louder**, **displacement**, **trough**, **midpoint**, **double**.

To measure the amplitude of a transverse wave, you need to measure the distance from the _____ (the equilibrium position) to a peak. This tells you how much

_____ the wave is carrying. We measure this in the direction of _____, so it looks slightly different for longitudinal and transverse waves, but is still the maximum _____ in both cases.

A common mistake is measuring all the way from a peak to a _____. Doing this will accidentally give you _____ the actual amplitude.

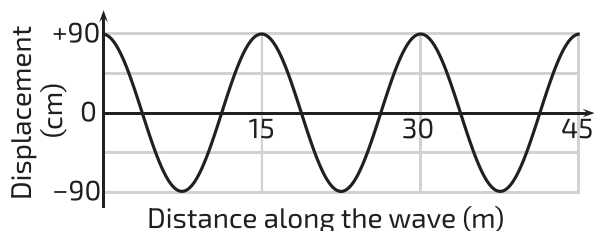
If you increase the amplitude of a sound wave, the sound will become _____, but the pitch will stay exactly the same.

- 5 Groggy the giant and Shirahoshi the giant mermaid live on a small planet called Newtonia.

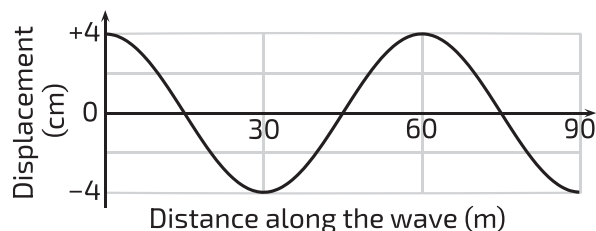
Like they do on earth, earthquakes on Newtonia create both longitudinal and transverse waves. Shirahoshi is famous for her incredibly loud voice - so much so that she can make the ground shake with her singing. Groggy says there is no way for him to detect the difference between Shirahoshi singing and a genuine earthquake.

Is he correct? Explain why.

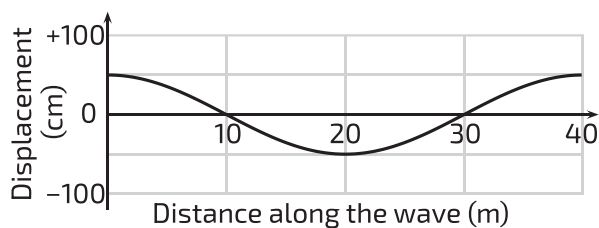
- 6 Write down the wavelength (λ) and the amplitude (A) of each wave.



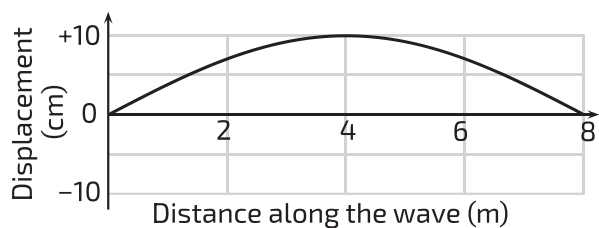
$\lambda =$ _____ $A =$ _____



$\lambda =$ _____ $A =$ _____



$\lambda =$ _____ $A =$ _____



$\lambda =$ _____ $A =$ _____

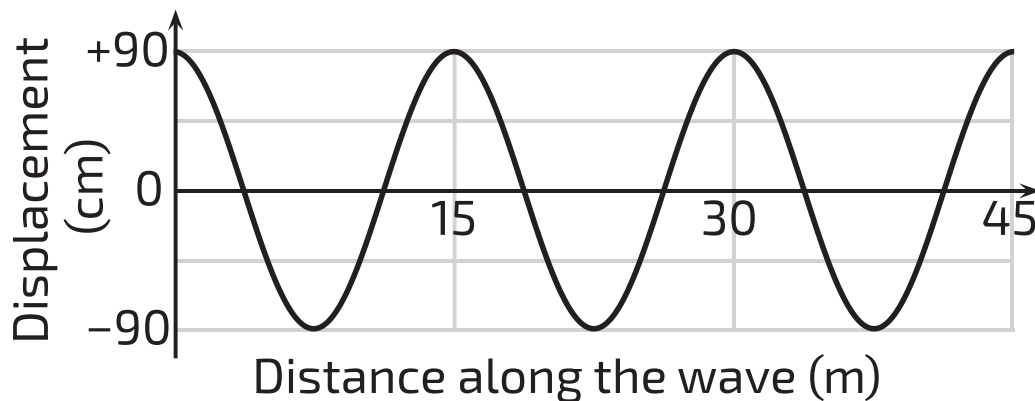
- 7 Which of these real world waves is the odd one out?

- A ripple moving across the surface of a puddle
- The sound of a teacher blowing a whistle
- A wave moving along a rope shaken up and down
- A "Mexican wave" travelling around a stadium crowd

8 Decide and label whether each of the waves below is transverse or longitudinal.

- Sound from a television
- A ripple on the surface of a lake
- A recently played guitar string
- The loud bang of a firework
- A vibrating spider web after a fly lands

9 Draw three sets of arrows showing lengths that someone could measure to measure exactly one wavelength. One arrow should be above the middle axis, one below the middle axis, and one *on* the middle axis



10 The giant mermaid Shirahoshi is singing underwater to send a message across the ocean to Groggy, who is on the shore.

To be safe, she thinks she should also write the message on paper, so a scroll is put in a bottle floating in the water directly in front of her mouth.

As she sings, her loud sound waves travel through the water and the bottle, towards the shore.

Which option(s) correctly describe the motion of the bottle?

- The bottle will stay in the same general area, bobbing back and forth in place
- The bottle vibrates because the water particles pass their energy forward without travelling far themselves
- The sound waves will push the bottle all the way to the shore where Groggy is standing
- The water particles around the bottle travel to the shore, leaving the bottle behind in a vacuum